# Quantum order by disorder among lozenge tilings: a staircase of hub-free crystals between the Lieb ferrimagnet and the antiferromagnet

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Lozenge tilings of the triangular lattice define an extensive family of bipartite networks, from the dice lattice, a Lieb ferrimagnet with the largest sublattice imbalance, to square-type tilings, which are compensated antiferromagnets. For the classical Heisenberg antiferromagnet all of them are degenerate. We show that quantum fluctuations lift this degeneracy through a local mechanism: to leading order the zero-point energy is a sum of single-site terms $\varepsilon(z)$ that are convex in the coordination $z$, so that Jensen's inequality at fixed $\langle z\rangle = 4$ selects the square-type tiling. A magnetic field, which couples to the Lieb moment, competes with this selection. Combining sign-free quantum Monte Carlo for $S = \frac{1}{2}$ to $\frac{3}{2}$ with an exhaustive enumeration of periodic tilings, we find that the competition produces a staircase of periodic crystals at $M_{\rm dice}/6$, $M_{\rm dice}/3$, $M_{\rm dice}/2$ and $2M_{\rm dice}/3$ instead of a single jump. The deepest crystal, at $M_{\rm dice}/2$ ($0.173J < h < 0.211J$ for $S = \frac{1}{2}$), contains no sixfold hubs, because the moment is carried most cheaply by pairs of threefold and fivefold sites. The low-field plateaus are described by a short-ranged lattice gas of curvature defects, and the step fields are independent of $S$ in units of $J$. In the height representation $z$ measures the discrete Gaussian curvature and the Lieb moment the sublattice-staggered curvature, so the field crumples an otherwise folded surface. We give spectroscopic signatures of each crystal and discuss realizations in nanographenes, atomic spin arrays and quantum simulators.

## I. INTRODUCTION

Order by disorder is the selection of a subset of classically degenerate ground states by fluctuations [1, 2]. In frustrated magnets the degenerate manifold is usually a family of spin configurations on a *fixed* lattice. Thermal or quantum fluctuations then pick the configurations with the softest excitation spectrum, most often collinear or coplanar ones. In this paper we study a different kind of degeneracy, one in which the object that is degenerate is the *lattice itself*.

Consider the lozenge (rhombus) tilings of the triangular lattice. Each tiling is obtained by pairing every up triangle with one of its down neighbors into a rhombus and deleting the shared edge. The remaining bonds form a planar bipartite graph in which every site has coordination $3 \le z \le 6$ and the mean coordination is exactly $\langle z\rangle = 4$ [Fig. 1]. Two extreme members of this family are familiar. The dice ($\mathcal{T}_3$) lattice [3, 4] has sixfold hubs and threefold rims. The square-type tiling has only fourfold sites and is topologically the square lattice. By Lieb's theorem [5], the Heisenberg antiferromagnet on any of these graphs has a ground state with total spin $S_{\rm tot} = S|N_A - N_B|$. The dice lattice is therefore a ferrimagnet with the largest possible moment $M_{\rm dice} = SN/3$, while the square-type tiling is compensated. The family thus spans ferrimagnets and antiferromagnets continuously.

For classical spins every tiling has the same energy, $E_{\rm cl} = -JS^2N_b = -2JS^2N$, because every tiling has $N_b = 2N$ bonds and collinear Néel order satisfies all of them. The number of tilings grows exponentially with $N$. Quantum mechanically the degeneracy is lifted, and two natural questions follow. Which lattice do quantum fluctuations select, and why? And how does a magnetic field, which couples directly to the Lieb moment and therefore favors the dice tiling, compete with that selection?

Quantum spins on dice-derived networks have been studied by Jagannathan, Wessel and co-workers [6–9]. They found that local "phason flips" of a staggered dice lattice raise the quantum ground-state energy and enhance the staggered moment [8], and that domain walls between phason domains carry a positive line energy ($\approx 0.062J/a$) and feel a Casimir-like $1/d$ interaction [9]. Their reference structure, the staggered dice lattice (SDL), is itself a lozenge tiling with coordinations $z = 3, 4, 5$ in equal numbers and $N_A = N_B$. The studies of phason flips kept the two sublattices balanced ($N_A = N_B$), considered random or dilute flips, and did not include a field. The questions posed above involve sectors with different sublattice imbalance, systematic control of that imbalance, and the competition with a Zeeman field. To our knowledge they have not been addressed.

Our main results are as follows. (i) The zero-point energy of any tiling is accurately a sum of single-site energies $\varepsilon(z)$, and $\varepsilon(z)$ is convex. Jensen's inequality therefore selects uniform $z = 4$ tilings, which as graphs are all square lattices, at zero field. The resulting selection energy is $\propto JS$ and can be computed for any tiling by counting coordinations [Fig. 2]. (ii) Domain walls between dice and square regions have *negative* energy, $\sigma \approx -0.013J/a$ [Fig. 3]. The field-driven crossover is therefore not a first-order jump through phase coexistence. (iii) The crossover is a magnetization staircase of periodic crystals, established by exhaustive enumera-

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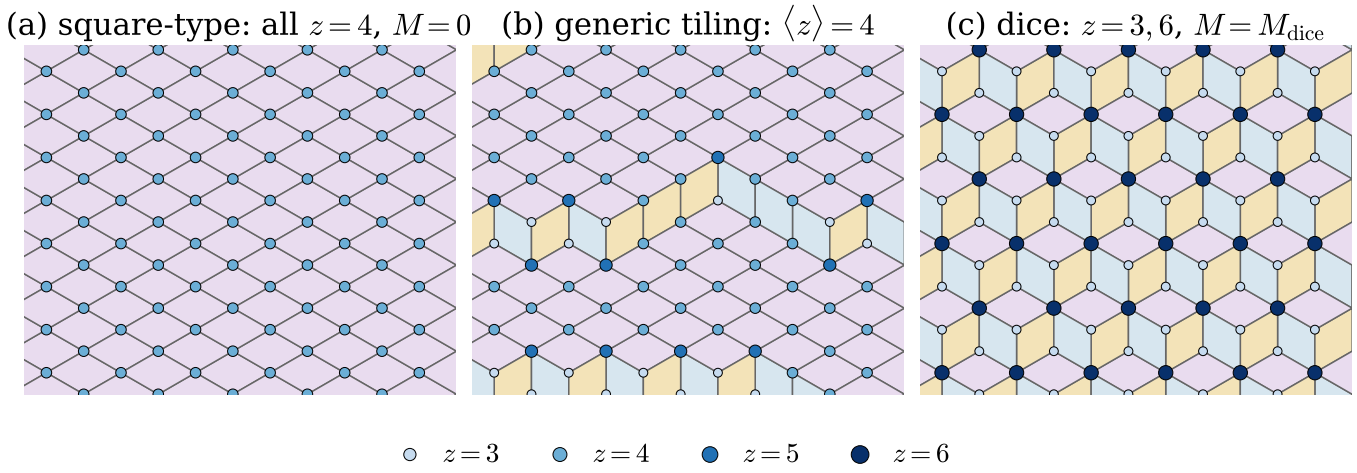



FIG. 1. Lozenge tilings of the triangular lattice and the bipartite networks they define. Rhombi are shaded by orientation, and sites are colored and sized by coordination $z$. (a) Square-type tiling: all $z = 4$, compensated ($M = 0$). (b) A generic tiling, with $z$ varying from site to site; every tiling has $\langle z\rangle = 4$. (c) Dice lattice: sixfold hubs and threefold rims, Lieb moment $M_{\rm dice} = SN/3$. All tilings have the same classical energy $-2JS^2N$.

tion and quantum Monte Carlo (QMC) [Figs. 4 and 5, Table I]. The deepest plateau crystal, at $M_{\rm dice}/2$, is hub-free, with $z = 3, 4, 5$ in equal numbers. Its neighbors at $M_{\rm dice}/6$, $M_{\rm dice}/3$ and $2M_{\rm dice}/3$ were resolved only with 36-site cells and $L = 36$ QMC; 48-site cells, screened with spin-wave theory, add no further step. Below $M_{\rm dice}/2$ the crystals are successive packing stages of $z = 3/z = 5$ dipoles, governed by a short-ranged defect interaction in which like-sign curvature attracts [Fig. 6]. (iv) Each plateau carries distinct fluctuation, entanglement and dynamical signatures [Figs. 7 and 8]. The plateau fields are $S$-independent in units of $J$ [Fig. 9]. In an effective tiling model the $M_{\rm dice}/2$ crystal melts at $T \simeq 0.02$–$0.03J$, and local-flip dynamics become trapped on cooling [Fig. 10]. (v) In the height representation, $z$ is exactly the discrete Gaussian curvature of a lifted surface, and the Lieb moment is its sublattice-staggered total. The results can thus be read geometrically: zero field selects folded, curvature-free surfaces, and the field crumples them with conical defects [Fig. 11].

The paper is organized as follows. Section II defines the model and its classical degeneracy, and Sec. III summarizes the methods. Section IV explains the zero-field selection. Section V presents the field-driven crystals and the staircase. Sections VI–VIII discuss fluctuations, the dependence on $S$, and finite temperature. Section IX gives the geometric (curvature) interpretation. Section X discusses interpretation, platforms and limitations, and Sec. XI concludes.

## II. MODEL AND CLASSICAL DEGENERACY

We consider

$$H_{\mathcal{T}} = J \sum_{\langle ij\rangle\in\mathcal{T}} \mathbf{S}_i \cdot \mathbf{S}_j - h \sum_i S_i^z, \qquad J > 0, \qquad (1)$$

where the sum runs over the bonds of a lozenge tiling $\mathcal{T}$ of an $L \times L$ triangular torus ($N = L^2$ sites) with primitive vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ at 60°. Each tiling is specified by a perfect matching of up and down triangles, which is equivalent to a dimer covering of the honeycomb lattice. Its height function has an average slope (tilt) that is conserved by the elementary hexagon flip, i.e., the rotation of three rhombi around a site with $z = 3$ [10]. The tilt therefore labels topological sectors on the torus.

Every tiling has $2N$ bonds, so $\langle z\rangle = 4$. The graph is bipartite with sublattices $A$ and $B$. By Lieb's theorem [5] the ground state has $S_{\rm tot} = S|N_A - N_B| \equiv M$. We write $m = |N_A - N_B|/(2N)$ for the imbalance per site, so that $m = 1/6$ for dice. The imbalance is not a topological invariant, since a single hexagon flip changes $N_A - N_B$ by 2. However, strips of dice and square-type tiling separated by straight walls provide a systematic family that interpolates between $m = 0$ and $m = 1/6$ at fixed tilt. We call it the "tilt family" below.

Classically, every tiling admits a collinear Néel state with energy $-JS^2$ per bond, so all tilings are degenerate at $h = 0$. For $h > 0$ the dice tiling wins immediately through its Zeeman energy $-hM$. The classical magnetization therefore jumps from 0 to $M_{\rm dice}$ at $h = 0^+$ [dashed line in Fig. 5(b)]. Everything that follows is a quantum effect.

## III. METHODS

*Quantum Monte Carlo.* All tilings considered here are bipartite, so Eq. (1) is sign-free. We use the stochastic series expansion [11] with directed loops [12] and heat-bath exit probabilities, for general $S$ and with the field included in the bond operators. (A deterministic switch-and-reverse loop does not change the parity of off-diagonal operators and can fail on graphs with odd cycles, as occur around defects. The heat-bath choice avoids this problem.) The code was checked against exact diagonalization (ED) of 27-site tori for $S = \frac{1}{2}, 1, \frac{3}{2}$. We use $\beta J = L$ for $S = \frac{1}{2}$ ($L = 12$–$36$) and $\beta JS = 12$ for the spin-$S$ runs at $L = 12$; energies are converged in $\beta$ within the statistical errors quoted. The zero-field energy of a tiling is obtained from runs at small $h$ as $e_0 = e + h(m_{\rm meas} + m)/2$. This linear interpolation of the Zeeman energy is exact for ferrimagnetic tilings, whose ground-state multiplet is pinned at $S_{\rm tot} = M$. For the compensated square-type tiling we use $\chi \simeq 0.065 J^{-1}$ per site for the uniform susceptibility. Our square-type energy, $e_0 = -0.66917(30)J$ at $L = 24$, is consistent with the square-lattice value [13]. As a graph the square-type tiling *is* the square lattice, so we use the thermodynamic-limit value $e_{\rm sq} = -0.669437(5)J$ of Ref. [13] as the zero-moment reference. This matters because finite-size corrections differ between compensated and ferrimagnetic tilings. In LSWT the energy per site of the compensated square-type tiling on an $L \times L$ torus deviates from the thermodynamic limit by $-1.2 \times 10^{-3}$, $-1.5 \times 10^{-4}$ and $-4 \times 10^{-5} J$ at $L = 12$, 24 and 36, whereas for the ferrimagnetic crystals the deviation is below $5 \times 10^{-6} J$ at $L = 36$ (e.g., $-4 \times 10^{-8} J$ for $M_{\rm dice}/2$). In particular, a size-dependent term proportional to the total moment, $\alpha|\mu|/N$ with $\mu = (N_A - N_B)/2$, which appears when a few uncompensated defects sit in a compensated background, is absent for crystals with an extensive moment. We therefore compare crystals at $L = 36$ with the thermodynamic square-lattice reference.

*Spin-wave theory on arbitrary graphs.* For a bipartite graph with adjacency matrix $A$ and degree matrix $D$, linear spin-wave theory (LSWT) gives magnon energies $\omega_\nu = JS|\lambda_\nu|$, where $\lambda_\nu$ are the eigenvalues of $\mathcal{L}^{1/2} P \mathcal{L}^{1/2}$, with $\mathcal{L} = D - A$ the graph Laplacian and $P = \mathrm{diag}(\eta_i)$ the sublattice signs ($\eta_i = \pm 1$). It also gives the ground-state energy $E = -JS^2 N_b + \frac{JS}{2}(\sum_\nu |\lambda_\nu| - \sum_i z_i)$. We diagonalize the full bosonic Hamiltonian with Colpa's method [14] to obtain spin deviations, Gaussian entanglement entropies and $S^{xx}(\mathbf{q}, \omega)$. Zero modes ($\omega < 10^{-2} J$) are excluded.

*Search over tilings.* We use three complementary searches. (a) *Periodic enumeration:* for every supercell $\mathbf{v}_1 = (p, 0)$, $\mathbf{v}_2 = (s, q)$ with $pq \le 48$ that is compatible with an $L = 36$ torus (67 cells), we enumerate all tilings (up to $2 \times 10^7$ per cell) and keep the bipartite ones (on a torus some tilings are not bipartite). We group them into structural classes by a translation-invariant fingerprint (the coordination histogram together with the multiset of neighbor coordinations) and keep one representative per class, retaining every class within $0.03J$ of the coordination-model minimum at each $m$, so that tilings with the same coordination histogram but different arrangements are kept. Candidates within $2 \times 10^{-3} J$ of the coordination-model minimum (Sec. IV) at each $m$, and every hub-free class for $pq \le 36$, are evaluated with LSWT in the thermodynamic limit, and the hull structures with QMC. (b) *Annealing:* simulated annealing in tiling space with local flips is performed at fixed tilt for $L = 18, 24, 30$, first with the coordination model and then with the full LSWT energy. This produces the irregular networks shown in gray in Fig. 5(a). (c) *Tilt family:* strips of dice and square tiling, used to extract wall energies.

*Finite-temperature tiling Monte Carlo.* To study thermal stability we treat the tiling as a classical degree of freedom with the coordination free energy of Sec. IV, scaled to QMC. We simulate it with local hexagon flips in the tilt sector of the crystal.

## IV. ZERO FIELD: WHY THE SQUARE-TYPE TILING WINS

### A. A local, convex zero-point energy

We first computed the LSWT energy of 75 tilings with widely different structure: tilt-family strips, and tilings with up to 200 random flips. Their energies are reproduced by a purely local form,

$$\frac{E_{\rm LSWT}}{N} \simeq \frac{1}{N} \sum_i \varepsilon(z_i) + c_m\, m, \tag{2}$$

with

$$\varepsilon(3, 4, 5, 6) = (-0.8544,\ -0.6582,\ -0.4423,\ -0.2061) J$$

and $c_m = 0.0304J$ for $S = \frac{1}{2}$. Because $\sum_i (z_i - 4) = 0$ for every tiling, $\varepsilon(z)$ is defined only up to a term $\lambda(z - 4)$, which changes no energy; the values above are the minimum-norm least-squares solution. Only $\varepsilon(4)$ and combinations that are invariant under this shift, such as the second differences and the costs of neutral groups below, are physical. The rms residual is $1.4 \times 10^{-4} J$, compared with a spread of energies of $6.8 \times 10^{-3} J$ [Fig. 2(b)]. The small $c_m$ term accounts for the weak cost of sublattice imbalance.

The key property is that $\varepsilon(z)$ is *convex* [Fig. 2(a)], a statement that is invariant under the shift above. Physically, the zero-point energy gained by spreading quantum fluctuations over more bonds saturates as $z$ grows, so replacing two $z = 4$ sites by a $z = 3$ and a $z = 5$ site costs energy. Because $\sum_i z_i = 4N$ is fixed for every tiling, Jensen's inequality gives

$$\frac{1}{N} \sum_i \varepsilon(z_i) \ \ge\ \varepsilon(\langle z \rangle) = \varepsilon(4), \tag{3}$$

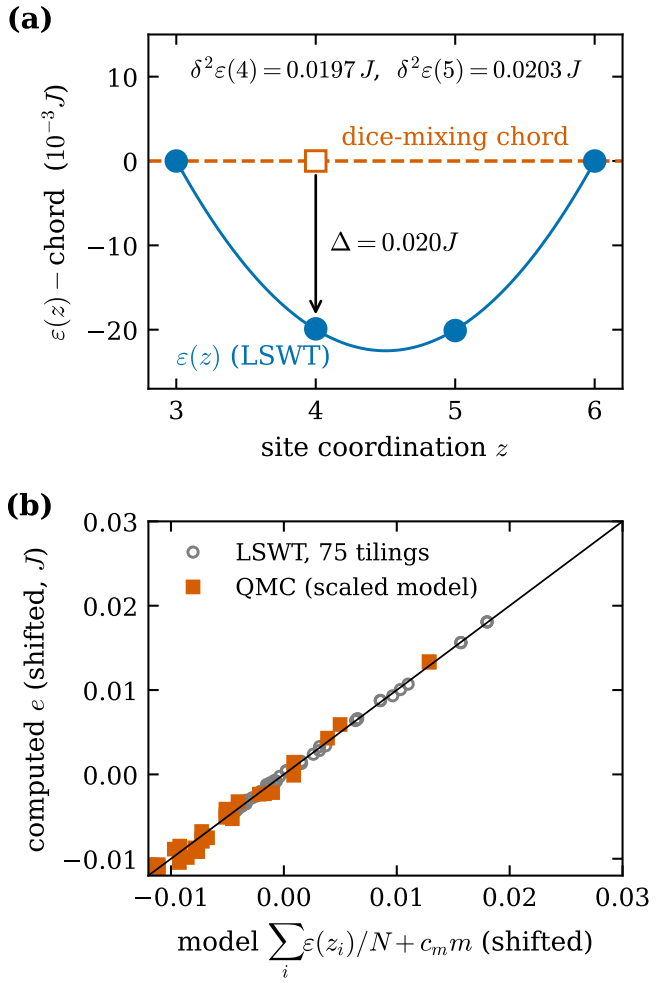



FIG. 2. Coordination mechanism of quantum tiling selection ($S = \frac{1}{2}$). (a) Single-site zero-point energies $\varepsilon(z)$ obtained by fitting LSWT energies of 75 tilings to Eq. (2), plotted relative to the chord joining $z = 3$ and $z = 6$ (dashed; the dice composition). The large linear trend of $\varepsilon(z)$ is removed, so its convexity appears as the dip below the chord. The discrete second differences $\delta^2\varepsilon(z) = \varepsilon(z+1) + \varepsilon(z-1) - 2\varepsilon(z)$ are $0.0197J$ ($z = 4$) and $0.0203J$ ($z = 5$). At fixed $\langle z\rangle = 4$ the uniform $z = 4$ tiling therefore lies below the dice combination (open square) by $\Delta = 0.020J$ per site. (b) Energies computed for many tilings vs the coordination model: LSWT (open circles, rms residual $1.4 \times 10^{-4}J$) and QMC with the model rescaled linearly (squares, rms $5 \times 10^{-4}$–$8 \times 10^{-4}J$). Both axes are shifted by a common constant.

with equality only when all $z_i = 4$. Every other tiling pays for its coordination variance. Tilings with all $z_i = 4$ are not unique: besides the square-type tiling there are folded variants (Sec. IX). As graphs, however, all of them are square lattices, so the zero-field ground state is in every case the square-lattice antiferromagnet. The dice lattice, whose coordinations $3, 3, 6$ lie farthest from 4, is the most expensive at $h = 0$.

*How good is the local model?* We tested Eq. (2) in four ways (Appendix A). (i) On an enlarged training set of 128 tilings ($L = 18, 24$, up to 600 random flips) the rms residual is $1.6 \times 10^{-4}J$ in leave-one-out cross-validation, with no systematic dependence on system size, number of flips or starting structure (largest class-averaged residual $1.8 \times 10^{-4}J$). (ii) Adding the number of bonds of each type $(z_i, z_j)$ as extra terms halves the residual, to $0.9 \times 10^{-4}J$. Correlations between neighboring coordinations are therefore present but small. (iii) Out of sample, on 71 periodic crystals evaluated with LSWT in the thermodynamic limit, the residual is $2.4 \times 10^{-4}J$ ($1.5 \times 10^{-4}J$ with pair terms). (iv) Tilings with *identical* coordination histograms but different arrangements differ by at most $3.5 \times 10^{-4}J$; these arrangement energies are described in Sec. V C. The model therefore resolves differences of order $10^{-3}J$, such as the depth of the $M_{\text{dice}}/2$ and $2M_{\text{dice}}/3$ hull points, but not differences of order $10^{-4}J$. We use it only to screen candidates. Every energy on which a conclusion rests is computed with QMC.

QMC confirms this picture quantitatively. For $S = \frac{1}{2}$ the square-type tiling has $e_0 = -0.669437(5)J$ [13] and the dice tiling $-0.63786(30)J$, a difference of $0.0316J$ per site, compared with zero classically. The QMC energies of all structures studied, including strips, periodic crystals and annealed networks, follow the coordination model after a single linear rescaling, $e_{\text{QMC}} = c_0 + c_1\, e_{\text{model}}$ with $c_1 = 1.27$, to $5$–$8 \times 10^{-4}J$ [Fig. 2(b)]. This model is what makes the exhaustive searches of Sec. V possible.

### B. The cost of a moment: why hubs are avoided

Equation (2) also tells us how a tiling can acquire a moment most cheaply. On any bipartite tiling each bond joins $A$ to $B$, so $\sum_{i\in A} z_i = \sum_{i\in B} z_i = 2N$. Writing $q_i = \pm(z_i - 4)$, with $+$ on the minority sublattice $A$ and $-$ on $B$, this gives

$$m = \frac{N_B - N_A}{2N} = \frac{1}{8N}\sum_i q_i. \tag{4}$$

Every site with $z \neq 4$ thus carries a "moment charge" $q_i$. Because $\sum_i (z_i - 4) = 0$, charges come in neutral groups, and only the cost of a neutral group is defined unambiguously. The cheapest group is a $z = 5$ site on $A$ with a $z = 3$ site on $B$: charge $+2$ at cost $\varepsilon(5) + \varepsilon(3) - 2\varepsilon(4) = 0.0197J$, i.e., $0.0099J$ per unit charge. A hub requires a $z = 6$ site on $A$ and two $z = 3$ sites on $B$: charge $+4$ at cost $\varepsilon(6) + 2\varepsilon(3) - 3\varepsilon(4) = 0.0597J$, i.e., $0.0149J$ per unit charge, 50% more. To leading order, the energy therefore rises *linearly* with $m$, with slope $8 \times 0.0099J + c_m \approx 0.11J$, as long as the moment can be carried by aligned 3/5 pairs alone. It rises more steeply once hubs become unavoidable.

The largest moment that 3/5 pairs can carry is fixed by the geometry of the tiling. In all 67 supercells with up to 48 sites that we enumerated (Sec. III), no hub-free tiling has $m > 1/12$, i.e., $M > M_{\text{dice}}/2$. (We retain every structural class within $0.03J$ of the model minimum at each

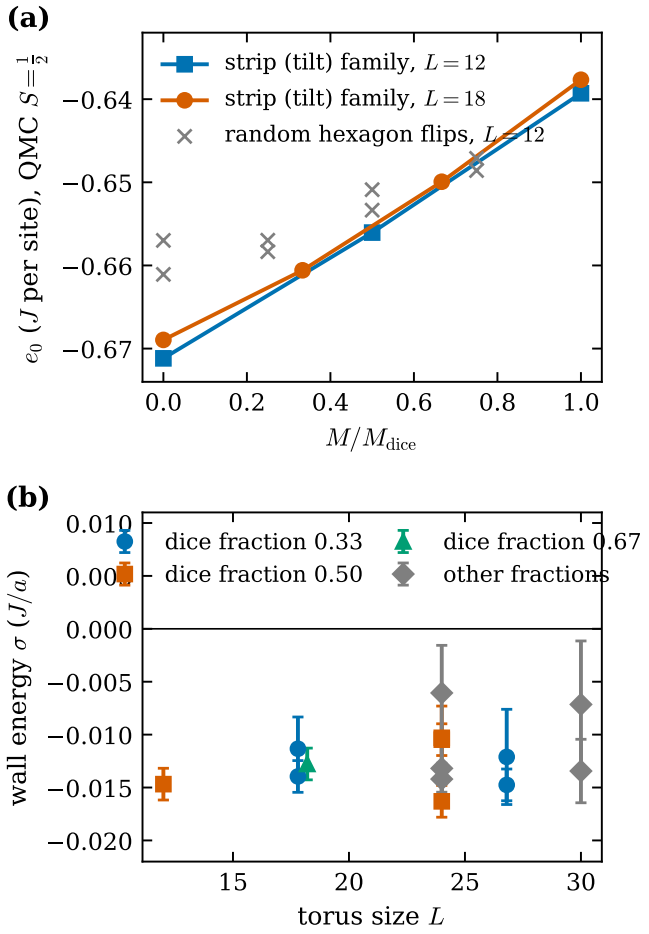



FIG. 3. (a) QMC ground-state energies ($S = \frac{1}{2}$) of the tilt family (parallel strips of dice and square tiling) for $L = 12$ and 18, together with tilings obtained by random hexagon flips (crosses). Random flips cost energy at fixed $M$; the classical energy is $-0.5J$ for all points. (b) Energy per unit length of a dice|square wall extracted from the strips, for several dice fractions and torus sizes. All values are negative.

$m$. A perfectly sorted hub-free tiling at larger $m$ would lie below every tiling with hubs at that $m$ and would have been retained.) The leading-order energy $e_{\rm model}(m)$ is therefore a straight line from the square-type tiling to the hub-free $M_{\rm dice}/2$ crystal, followed by a steeper segment to dice. The $M_{\rm dice}/2$ crystal sits at the kink between them. This gives two predictions, both borne out below. The $M_{\rm dice}/2$ plateau is robust, because it rests on the leading term. Structures with $0 < M < M_{\rm dice}/2$ are nearly degenerate at leading order, so the low-field plateaus are decided by smaller arrangement-dependent energies, which we quantify in Sec. V C.

### C. Tilt family and negative walls

Figure 3(a) shows the QMC energy of the tilt family as a function of its moment. The curve is slightly convex: intermediate strip patterns lie *below* the straight line joining the square-type and dice tilings. Subtracting the bulk energies gives the energy per unit length of a dice|square wall. It is negative for every width, fraction and system size studied, $\sigma \approx -0.011$ to $-0.015J/a$ [Fig. 3(b)]. The coordination picture explains the sign. A wall converts rim ($z = 3$) and hub ($z = 6$) sites of dice into $z = 4, 5$ sites, which lowers $\sum_i \varepsilon(z_i)$ because $\varepsilon$ is convex. By contrast, random flips at fixed $M$ (crosses) always raise the energy, in agreement with Ref. [8]. The sign of $\sigma$ is also opposite to that of the phason-flip walls inside the SDL, whose line energy is $+0.062J/a$ [9]. Those walls separate two tilings with the same coordination histogram, so they cannot lower $\sum_i \varepsilon(z_i)$; a dice|square wall can.

A negative wall energy has a direct consequence. When the field makes dice favorable, the system does not phase-separate into large dice and square domains. It instead gains energy by creating walls. In LSWT on $L = 36$ and 48 tori the energy per straight wall is $\sigma_\infty \approx -0.0088J/a$ for widely separated walls. It becomes about $0.0005J/a$ more negative when the walls are six rows apart, the closest spacing compatible with bipartiteness [Fig. 12(b)]. Straight walls therefore attract at short distance. The wall picture explains why intermediate structures win, but it does not select the crystal. Superlattices of straight walls lie 2.0–2.5 $\times 10^{-3}J$ above the best crystals at the same $m$ (LSWT at $m = 1/18$, $1/12$ and $1/9$). The reason is their histogram: the straight-wall superlattice at $M_{\rm dice}/2$ still contains 8% hubs, whereas the crystal has none. The selection is controlled by the coordination histogram, as in Sec. IV B, and the negative wall energy is its consequence rather than an independent cause.

## V. FIELD-DRIVEN CRYSTALS AND THE MAGNETIZATION STAIRCASE

### A. Exhaustive search of periodic tilings

Because the energy of a tiling is essentially a function of its coordination histogram, we can scan the complete set of periodic tilings with small unit cells. For all 67 supercells with up to 48 sites that fit an $L = 36$ torus we enumerated every tiling (2.6 $\times 10^5$–1.9 $\times 10^7$ per cell for the 36- and 48-site cells) and retained the bipartite ones, one representative per structural class (Sec. III). The candidates were ranked with LSWT in the thermodynamic limit (a Bloch calculation on the supercell, which agrees with $L = 36$ tori to $10^{-8}J$), and the hull structures were evaluated with QMC. In a 36-site cell $N_A - N_B$ is even, so $M/M_{\rm dice} = k/6$; the 48-site cells add $M/M_{\rm dice} = k/8$. Enlarging the cells to 36 sites uncovered the $M_{\rm dice}/6$ and $2M_{\rm dice}/3$ crystals. An earlier version of this search kept only the lowest few tilings per $m$ according to Eq. (2). Because that model depends only on the histogram, it discarded tilings with the same histogram but a better arrangement. The class-resolved search found $M_{\rm dice}/6$ and $M_{\rm dice}/3$ structures that are 2.5 and 1.1$\times 10^{-4}J$ lower

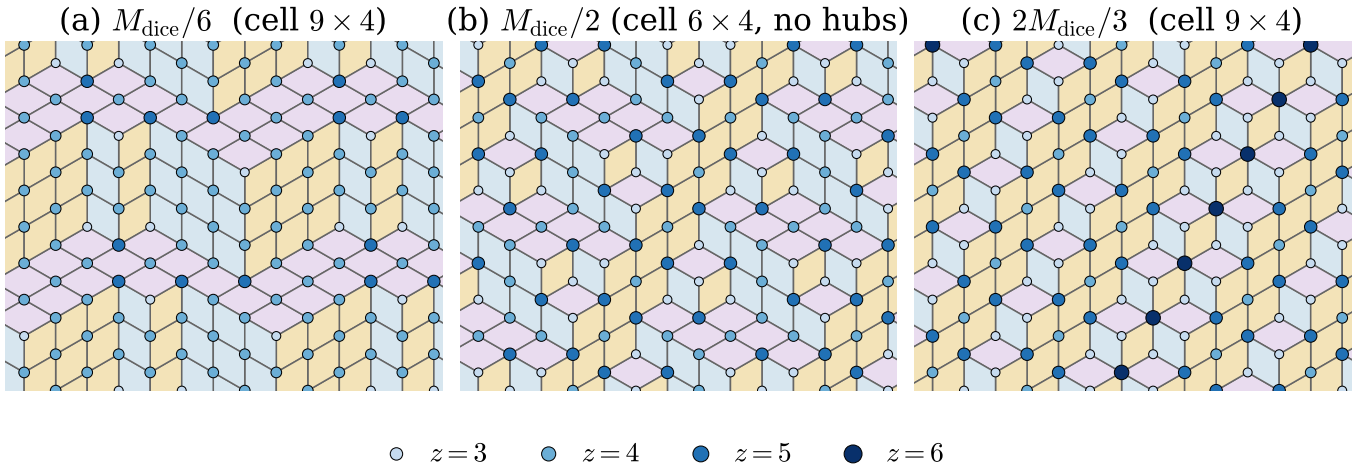



FIG. 4. Three of the periodic tilings on the lower convex hull of $e_0(m)$ (QMC, $S=\frac{1}{2}$); the $M_{\rm dice}/3$ crystal is shown in Fig. 11(c). (a) $M_{\rm dice}/6$ crystal (cell $9\times4$, shear $s=4$): dilute $z=3/z=5$ dipoles in a square-type background. (b) $M_{\rm dice}/2$ crystal (cell $6\times4$, $s=4$): $z=3,4,5$ in equal numbers and *no* sixfold hubs. (c) $2M_{\rm dice}/3$ crystal (cell $9\times4$, $s=7$): the first appearance of hubs, $z_3{:}z_4{:}z_5{:}z_6=16{:}6{:}12{:}2$. Color code as in Fig. 1.

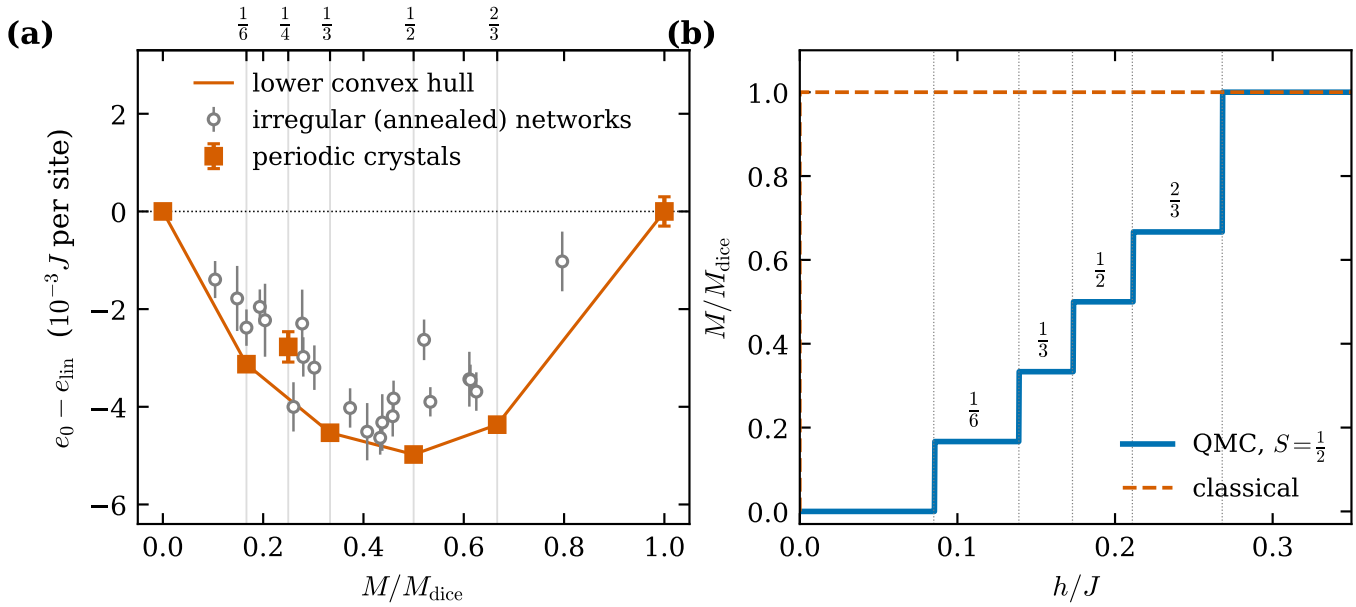



FIG. 5. Magnetization staircase ($S=\frac{1}{2}$, QMC). (a) Zero-field energy relative to the straight line between the square-type and dice tilings, $e_{\rm lin}(m)=e_{\rm sq}+6m\,(e_{\rm dice}-e_{\rm sq})$. Squares: best periodic crystal at each $M/M_{\rm dice}$ (cells $\le 36$ sites; $L=36$ for $\frac{1}{6},\frac{1}{3},\frac{1}{2},\frac{2}{3}$, $L=24$ for $\frac{1}{4}$). Circles: irregular networks from LSWT annealing at $L=18,24,30$. The line is the lower convex hull. (b) $T=0$ magnetization obtained by minimizing $e_0-hm$ over the hull (solid), compared with the classical jump (dashed). Dotted vertical lines mark the step fields listed in Table I.

in LSWT, and QMC confirms both (Table I). The 48-site cells add no hull point: the best $M_{\rm dice}/4$ and $3M_{\rm dice}/8$ structures lie 1.0 and $0.6\times10^{-4}J$ above the LSWT hull, and those at $5M_{\rm dice}/8$, $3M_{\rm dice}/4$ and $7M_{\rm dice}/8$ lie well above it.

Figure 4 shows the crystals that lie on the lower convex hull, and Fig. 5(a) and Table I give their energies. Three features stand out.

(1) The deepest point of the hull is the $M_{\rm dice}/2$ crystal, $\delta e=-4.98(9)\times10^{-3}J$ at $L=36$ [$-5.3(4)\times10^{-3}J$ at $L=24$]. Remarkably, it contains *no* hubs. Its sites have $z=3,4,5$ in equal numbers. The crystal is thus the closest to the dice moment that we find while keeping every coordination adjacent to 4, which is exactly what a convex $\varepsilon(z)$ rewards. It has the same coordination histogram as the compensated SDL of Refs. [8, 9]. The difference is the arrangement: in the $M_{\rm dice}/2$ crystal the arrangement places the surplus sites on one sublat-

TABLE I. Periodic crystals on or near the lower hull ($S = \frac{1}{2}$). $M/M_{\rm dice} = 6m$. Coordination counts are per cell. $\delta e = e_0 - e_{\rm lin}$ from QMC, per site, with $L$ the torus size used ($\beta J = L$); the square-type reference is the thermodynamic-limit value [13]. The last column gives the field window in which the crystal is the $T = 0$ ground state.

| $M/M_{\rm dice}$ | cell $(p, q, s)$ | $z_3$:$z_4$:$z_5$:$z_6$ | $\delta e$ $(10^{-3}J)$ | $L$ | window $(h/J)$ |
|---|---|---|---|---|---|
| 0 | square | 0:1:0:0 | 0 | $\infty$ | $< 0.085$ |
| $\frac{1}{6}$ | $(9, 4, 4)$ | 4:28:4:0 | $-3.13(11)$ | 36 | 0.085–0.139[a] |
| $\frac{1}{4}$ | $(4, 6, 0)$ | 5:14:5:0 | $-2.77(31)$ | 24 | — |
| $\frac{1}{3}$ | $(9, 4, 4)$ | 8:20:8:0 | $-4.53(9)$ | 36 | 0.139–0.173[b] |
| $\frac{1}{2}$ | $(6, 4, 4)$ | 8:8:8:0 | $-4.98(9)$ | 36 | 0.173–0.211 |
| $\frac{2}{3}$ | $(9, 4, 7)$ | 16:6:12:2 | $-4.37(12)$ | 36 | 0.211–0.268 |
| 1 | dice | 2:0:0:1 | 0 | 24 | $> 0.268$ |

[a] Mean of three independent runs (random seeds). The structure used before the class-resolved search, cell $(9, 4, 5)$, gives $-2.68(10)$.
[b] Mean of two independent runs. The structure used before, cell $(3, 12, 0)$, gives $-4.39(13)$ at $L = 36$ and $-3.6(3)$ at $L = 24$; a third structure gives $-3.88(53)$ at $L = 18$.

tice, giving $m = 1/12$ instead of 0. The comparison also provides an independent check of Eq. (2). For the SDL it predicts $E_{\rm LSWT}/N = [\varepsilon(3) + \varepsilon(4) + \varepsilon(5)]/3 = -0.6516J$, in agreement with the value $-0.6517(7)J$ obtained by direct LSWT in Ref. [9]. The QMC energy of the SDL, $-0.6639J$ [8], lies $5.5 \times 10^{-3}J$ above the square-lattice value, again as Jensen's inequality requires.

(2) The $2M_{\rm dice}/3$ crystal, $\delta e = -4.37(12) \times 10^{-3}J$, lies $1.0 \times 10^{-3}J$ below the chord between the $M_{\rm dice}/2$ crystal and dice, about six combined standard errors. It is the first hull structure that contains hubs, but only one site in 18.

(3) At low moment the $M_{\rm dice}/6$ crystal, a dilute array of $z = 3/z = 5$ dipoles in a square-type background, lies $0.9 \times 10^{-3}J$ below the chord from $m = 0$ to $M_{\rm dice}/3$. The $M_{\rm dice}/3$ crystal, a pattern of alternating $z = 3/z = 5$ chains, lies $0.47 \times 10^{-3}J$ below the chord between $M_{\rm dice}/6$ and $M_{\rm dice}/2$. The $M_{\rm dice}/4$ structure is above the hull.

Irregular networks produced by annealing (gray circles) never reach the hull. They come from 55 LSWT-driven annealing runs at $L = 18$ and 106 two-stage (model, then LSWT) runs at $L = 24$ and 30. The runs spanned five tilt sectors, fields 0–$0.3J$ and two to three random seeds per setting. The best of them, near $M/M_{\rm dice} \approx 0.4$–$0.5$, have $\delta e \approx -4.4 \times 10^{-3}J$, about $0.6 \times 10^{-3}J$ above the $M_{\rm dice}/2$ crystal. As anticipated, the ground state in a field is a symmetric crystal, not a disordered network.

### B. Staircase

At $T = 0$ the tiling selected by a field $h$ minimizes $e_0(m) - hm$ (for $S = \frac{1}{2}$, where the Zeeman energy per site is $-hS_{\rm tot}/N = -hm$), with $-\frac{1}{2}\chi h^2$ added for the compensated tiling. Minimizing over the hull yields the staircase of Fig. 5(b):

$$0 \xrightarrow{0.085} \tfrac{1}{6} \xrightarrow{0.139} \tfrac{1}{3} \xrightarrow{0.173} \tfrac{1}{2} \xrightarrow{0.211} \tfrac{2}{3} \xrightarrow{0.268} 1,$$

with step fields in units of $J$ and magnetization in units of $M_{\rm dice}$. The classical jump at $h = 0^+$ is thus replaced by four plateaus between $h = 0.085J$ and $0.268J$, of widths $0.054$, $0.034$, $0.038$ and $0.057J$. This staircase is established within the searched space: periodic cells with up to 36 sites (QMC) and 48 sites (LSWT), plus the annealed networks. It is not a proof over all tilings. The deepest point of the hull is the hub-free $M_{\rm dice}/2$ crystal. The widest plateaus are those of $M_{\rm dice}/6$ and of $2M_{\rm dice}/3$, which has a hub on one site in 18. Each step corresponds to a first-order change of the *lattice* and not only of the spin state.

*Robustness.* We assess the staircase in three ways. (i) *Statistical errors.* Resampling all QMC energies within their errors (4000 samples, Table II) keeps every plateau, $M_{\rm dice}/6$, $M_{\rm dice}/3$, $M_{\rm dice}/2$ and $2M_{\rm dice}/3$, in essentially all samples, while $M_{\rm dice}/4$ almost never appears. The least secure plateau is $M_{\rm dice}/3$, whose margin is $0.47 \times 10^{-3}J$. With the structure used previously, its $\delta e$ depended on system size ($-4.39(13) \times 10^{-3}J$ at $L = 36$ but $-3.6(3) \times 10^{-3}J$ at $L = 24$), and the improved structure has so far been evaluated only at $L = 36$. (ii) *System size.* The $M_{\rm dice}/2$ deviation is $-5.3(4)$ and $-4.98(9) \times 10^{-3}J$ at $L = 24$ and 36, and $-4.8 \times 10^{-3}J$ at $L = 12$ relative to $L = 12$ references (Table III). The dice reference agrees between $L = 18$ and 24 to $10^{-4}J$, and the square-type reference is taken in the thermodynamic limit (Sec. III). In LSWT the crystal energies at $L = 36$ coincide with the thermodynamic limit. (iii) *Cell size.* With the class-resolved enumeration the best $M_{\rm dice}/2$ structure appears in 24-site cells and the best $M_{\rm dice}/6$ and $M_{\rm dice}/3$ structures in 36-site cells, and the 48-site cells lower no hull point. The $2M_{\rm dice}/3$ optimum first appears in 36-site cells, where its energy has dropped by $1.9 \times 10^{-3}J$ relative to 18-site cells; 48-site cells cannot host $M/M_{\rm dice} = \frac{2}{3}$, so its convergence is not demonstrated. Larger cells may still add steps nearby. Lowering an existing hull point cannot remove it from the hull, but a new structure at a nearby $m$ could. The leading-order analysis of Sec. IV B suggests that the

TABLE II. Robustness of the staircase to the QMC statistical errors: fraction of 4000 resampled hulls in which each plateau appears, and mean and standard deviation of the step fields (in $J$).

| plateau $M/M_{\rm dice}$ | present | lower edge $h/J$ |
|---|---|---|
| $\frac{1}{6}$ | 1.00 | $0.085 \pm 0.005$ |
| $\frac{1}{4}$ | 0.00 | — |
| $\frac{1}{3}$ | 1.00 | $0.139 \pm 0.005$ |
| $\frac{1}{2}$ | 1.00 | $0.173 \pm 0.005$ |
| $\frac{2}{3}$ | 1.00 | $0.211 \pm 0.006$ |
| 1 (dice) | 1.00 | $0.268 \pm 0.006$ |

$M_{\rm dice}/2$ kink is protected, and the short range of the defect interaction (Sec. V C) suggests that any additional steps below $M_{\rm dice}/2$ are very narrow.

Lieb's theorem fixes $S_{\rm tot} = M$ at $h = 0$. Because $S^z_{\rm tot}$ is conserved on a fixed graph and states with larger $S^z_{\rm tot}$ are separated by a magnon gap, the spin ground state stays pinned at $S^z_{\rm tot} = M$ throughout each plateau.

### C. A lattice gas of curvature defects

The site model treats every $z \neq 4$ site independently and misses the arrangement energies that decide the low-field plateaus; on hub-free tilings its residual is $2.7 \times 10^{-4} J$. We therefore describe a hub-free tiling as a lattice gas of curvature defects, the sites with $k_i = 4 - z_i = \pm 1$, each on a sublattice $\eta_i = \pm 1$, and fit

$$\frac{E_{\rm LSWT}}{N} = e_{\rm sq} + \varepsilon_d\, \rho_d + c'_m\, m + \frac{1}{N} \sum_{i<j} W(r_{ij}; k_i k_j, \eta_i \eta_j), \tag{5}$$

where $\rho_d$ is the defect density. The pair energy $W$ depends on the distance and on whether the two defects have the same curvature and lie on the same sublattice, and bonded and unbonded nearest neighbors are distinguished. We fit Eq. (5) to the Bloch-LSWT energies of all 4510 structurally distinct hub-free tilings with cells of up to 36 sites, keeping pairs up to $r = \sqrt{7}$ [extending the cutoff to $r = 4$, Fig. 6(b), does not lower the residual]. The cross-validated residual is $1.5 \times 10^{-5} J$, 18 times smaller than for the site model [Fig. 6(a)]. On 74 hub-free 48-site structures that were not used in the fit it is $2.1 \times 10^{-5} J$. The fitted parameters are $\varepsilon_d = 6.92(2) \times 10^{-3} J$ and $c'_m = 1.99(3) \times 10^{-3} J$, and three rules follow from the pair energies.

*(1) Sign.* In 14 of the 16 pair channels sign $W = -k_i k_j$, i.e., defects of the same curvature attract and defects of opposite curvature repel; the two exceptions are within the resolution. Bonded nearest neighbors dominate, with $W = -3.24(2) \times 10^{-3} J$ for like and $+4.39(2) \times 10^{-3} J$ for unlike pairs. The field, which couples to the staggered curvature [Eq. (7)], is what sorts the curvature onto the two sublattices (Sec. IX); the pair energies then decide how the sorted defects are arranged.

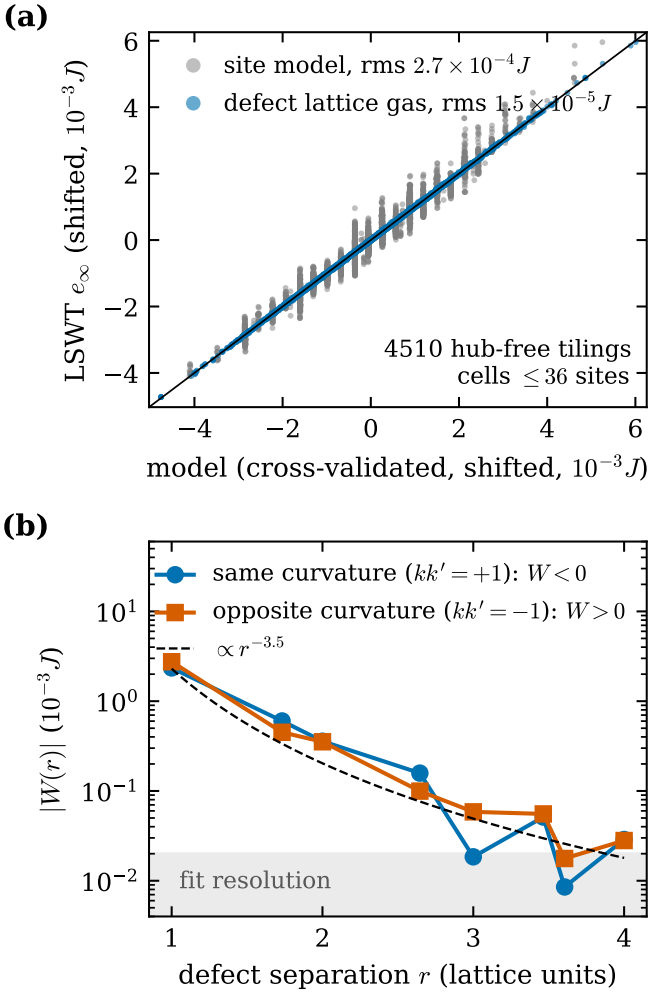



FIG. 6. Defect lattice gas, Eq. (5), fitted to thermodynamic-limit LSWT energies ($S = \frac{1}{2}$) of all structurally distinct hub-free periodic tilings with cells of up to 36 sites. (a) Cross-validated prediction vs LSWT: site model, Eq. (2), refitted to the same set (gray), and the lattice gas with pair terms up to $r = \sqrt{7}$ (blue). (b) Magnitude of the pair energy $W$ for defects of the same (circles) and opposite (squares) curvature, averaged over the two sublattice relations; bonded nearest neighbors are included at $r = 1$. Same-curvature pairs attract and opposite-curvature pairs repel. The dashed line is $\propto r^{-3.5}$; the shaded band marks the resolution of the fit.

*(2) Range.* $|W|$ decreases by a factor of about 30 between $r = 1$ and $r = 3$ [Fig. 6(b)]. A power law $\propto r^{-3.5}$ describes the four resolved shells better than an exponential, and beyond $r \approx 3$ the couplings are below $2 \times 10^{-5} J$. Resolving $W$ into a curvature channel ($\propto k_i k_j$) and a moment channel ($\propto q_i q_j$, with $q_i = -\eta_i k_i$ the moment charge of Eq. (4)) shows that the curvature channel dominates at short range, while at $r = 2$ and $\sqrt{7}$ the moment channel is comparable and repulsive between equal moments ($+0.10$ and $+0.08 \times 10^{-3} J$). This is the sign and power law expected for the magnon-mediated interaction

between uncompensated moments, which decays as $r^{-3}$ for phason flips in the staggered dice lattice [8]. In a polarized crystal all moments are equal, so this tail is repulsive and further favors evenly spaced dipoles.

*(3) Packing.* In every hull crystal each defect has at least one bonded neighbor of opposite curvature, so the elementary object is a $z = 3/z = 5$ dipole. Along the hull the number of such contacts per defect increases in steps, from 1.00 (mostly isolated dipoles, $M_{\rm dice}/6$) to 2.00 (alternating chains, $M_{\rm dice}/3$) and 2.75 (touching chains, $M_{\rm dice}/2$), while the number of unbonded nearest-neighbor contacts between like-curvature defects increases from 0.25 to 0.50 and 1.00. Each low-field plateau is thus a packing stage of dipoles, and each step field is set by the cost of the next contact divided by the moment it adds. Within each $m$, the model ranks the polarized structures in the same order as LSWT (Spearman correlation 0.93–0.98) and picks the same lowest structure.

The model also gives the onset of the staircase. An isolated, field-aligned dipole costs $E_{\rm dip} = 2\varepsilon_d + W_1^{\rm unlike} + c'_m/4 = 18.7 \times 10^{-3} J$ and carries $mN = \frac{1}{4}$, so the first step is expected at $h \approx 4E_{\rm dip} = 0.075J$. This agrees with the LSWT hull ($0.073J$). QMC gives $0.085 \pm 0.005J$, larger by a factor of 1.16, somewhat smaller than the factors 1.3–1.4 found for the other steps (Appendix B). Finally, because $W$ is short ranged, structures between the hull crystals differ from them by mixed contact numbers, which costs $\gtrsim 0.6 \times 10^{-4} J$ ($M_{\rm dice}/4$ and $3M_{\rm dice}/8$ above). Further steps would require couplings beyond $r \approx 3$ that exceed the resolution of $2 \times 10^{-5} J$, which would give field windows of order $10^{-3} J$ at most. The model is fitted to LSWT and covers only hub-free tilings, so it cannot exclude such steps, but it explains why the visible staircase is short.

## VI. FLUCTUATIONS, ENTANGLEMENT AND SPECTROSCOPIC SIGNATURES

Because the spin state within a plateau does not change with $h$, the quantum-fluctuation content of the ground state changes only at the steps, where the lattice changes. Figure 7 follows three measures along the staircase. The mean spin deviation decreases from 0.173 (square-type) through 0.172, 0.155, 0.140 and 0.124 to 0.095 (dice), and the single-site entanglement entropy falls from 0.49 to 0.32. The half-system entanglement per unit boundary length first rises slightly at the $M_{\rm dice}/6$ step and then drops by 28% between the square-type and dice tilings. The field thus reduces quantum fluctuations in discrete steps, by selecting lattices with more $z = 3$ sites on the majority sublattice, whose moments are the most nearly classical (their mean spin deviation falls from 0.16 in the $M_{\rm dice}/6$ crystal to 0.07 in dice). The values refer to $L = 36$ with zero modes excluded; the thermodynamic LSWT value for the square lattice is 0.197. This is the mirror image of the zero-field selection.

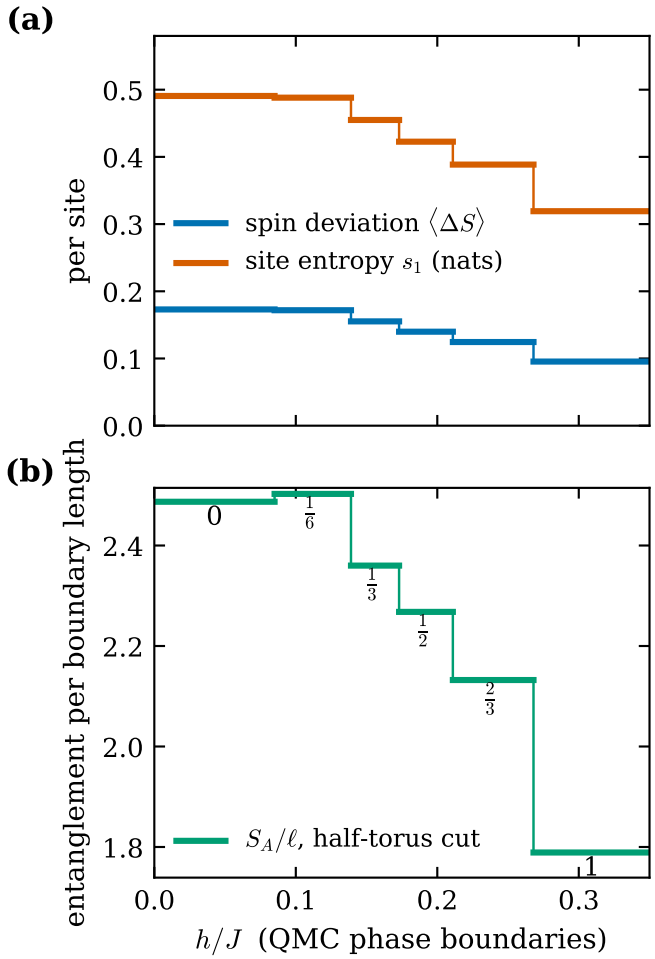



FIG. 7. Quantum-fluctuation measures of the LSWT ground state ($S = \frac{1}{2}$, $L = 36$) of the crystal selected at each field, using the QMC phase boundaries. (a) Mean spin deviation $\langle \Delta S \rangle = \langle a^\dagger a \rangle$ and single-site entanglement entropy. (b) Gaussian entanglement entropy of half the torus per unit boundary length. Plateau labels give $M/M_{\rm dice}$.

The crystals can be distinguished spectroscopically [Fig. 8]. The square-type tiling shows a single linearly dispersing magnon branch. The dice tiling combines a Goldstone branch with the dispersionless band near $\omega \approx 1.5J$ localized on the rims [3, 4]. The $M_{\rm dice}/2$ crystal shows backfolded minibands with small gaps at the zone boundaries of its $6 \times 4$ cell. Its low-energy branch is square-like, and its upper spectral weight is spread over a band of nearly flat modes. These features give a direct fingerprint of each plateau in inelastic neutron scattering or in magnon spectroscopy of engineered structures.

## VII. DEPENDENCE ON SPIN LENGTH

Classically the selection energy vanishes, and in spin-wave theory it is of order $JS$, whereas the exchange energy is of order $JS^2$. QMC for $S = \frac{1}{2}$, 1 and $\frac{3}{2}$

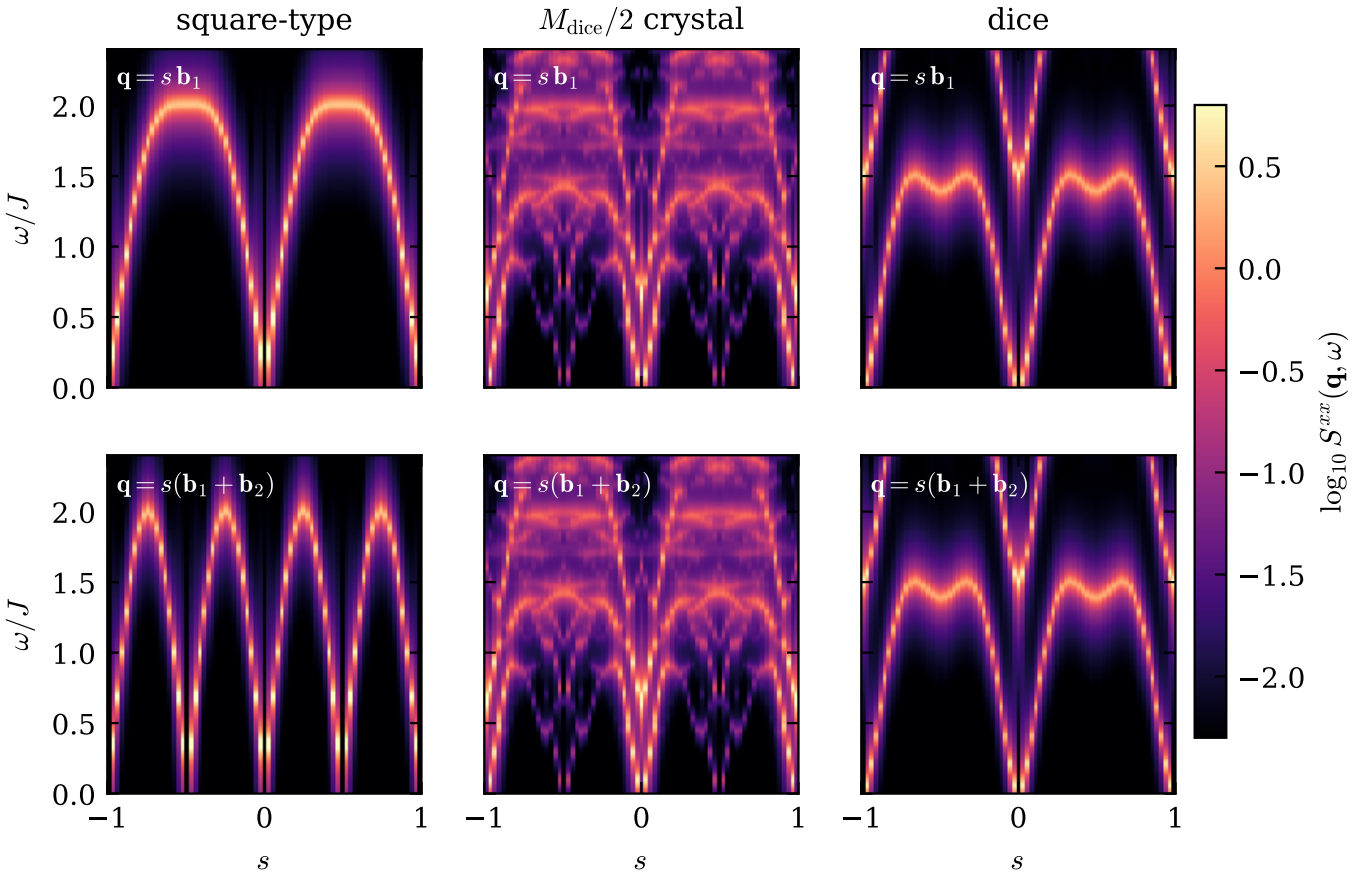



FIG. 8. LSWT transverse dynamical structure factor $\log_{10} S^{xx}(\mathbf{q},\omega)$ ($S = \frac{1}{2}$, $L = 36$, Lorentzian broadening $0.03J$) along $\mathbf{q} = s\,\mathbf{b}_1$ (top) and $\mathbf{q} = s(\mathbf{b}_1 + \mathbf{b}_2)$ (bottom), where $\mathbf{b}_{1,2}$ are the reciprocal vectors of the triangular lattice. Left: square-type tiling (a single sharp magnon branch). Center: $M_{\rm dice}/2$ crystal (backfolded minibands and gaps set by the $6 \times 4$ cell). Right: dice (a Goldstone branch plus the dispersionless band near $1.5J$ that comes from the rim sites).

confirms this scaling (Fig. 9 and Table III). The ratios $(e_{\rm dice} - e_{\rm sq})/S$ and $\delta e(M_{\rm dice}/2)/S$ are nearly constant and consistent with the LSWT values. Both the energy differences and the Zeeman energy $hM \propto hS$ are linear in $S$, so the step fields are $S$-independent in units of $J$. The field at which the $M_{\rm dice}/2$ crystal gives way to dice is $0.22$–$0.25J$ for all three spins. Measured against the saturation field of order $12JS$, the intermediate region therefore shrinks as $1/S$. The staircase is a genuinely quantum phenomenon, most pronounced for $S = \frac{1}{2}$; within our $L = 12$ data, the $M_{\rm dice}/2$ →dice field does not depend on $S$ in units of $J$.

TABLE III. Spin-$S$ dependence (QMC, $L = 12$). Energies per site in units of $J$.

| $S$ | $(e_{\rm dice} - e_{\rm sq})/S$ | $\delta e(M_{\rm dice}/2)/S$ | $h_c(M_{\rm dice}/2 \to \text{dice})$ |
|---|---|---|---|
| $\frac{1}{2}$ | 0.065 | −0.0097 | 0.25 |
| 1 | 0.058 | −0.0079 | 0.22 |
| $\frac{3}{2}$ | 0.059 | −0.0086 | 0.23 |
| LSWT | 0.050 | −0.0077 | — |

## VIII. FINITE TEMPERATURE IN AN EFFECTIVE TILING MODEL

When the tiling can rearrange, for example during growth, self-assembly or in a programmable array, the plateau crystals compete with the configurational entropy of the tiling. We model this with the QMC-scaled coordination free energy and local hexagon flips in the crystal's tilt sector. The spins are not sampled thermally. The temperatures below are therefore melting scales of this effective model, not transition temperatures of the full spin-plus-tiling system. Heating the $M_{\rm dice}/2$ crystal at fixed field [Fig. 10(a)] produces a sharp loss of crystalline overlap at $T_m = 0.020$, $0.025$ and $0.030J$ for $h = 0.15$, $0.19$ and $0.23J$. The results for $L = 24$ and 48 coincide. $T_m$ increases with $h$ [Fig. 10(b)] because the field deepens the hull minimum at $M_{\rm dice}/2$. In this effective model, whose hull has only the $M_{\rm dice}/2$ kink, the $M_{\rm dice}/2$ crystal is the ground state for $h \approx 0.13$–$0.25J$; all three fields lie in this window, although the outer two lie outside the narrower QMC window of Table I. On slow cooling from the disordered state, however, the system does not recrystallize. The overlap remains

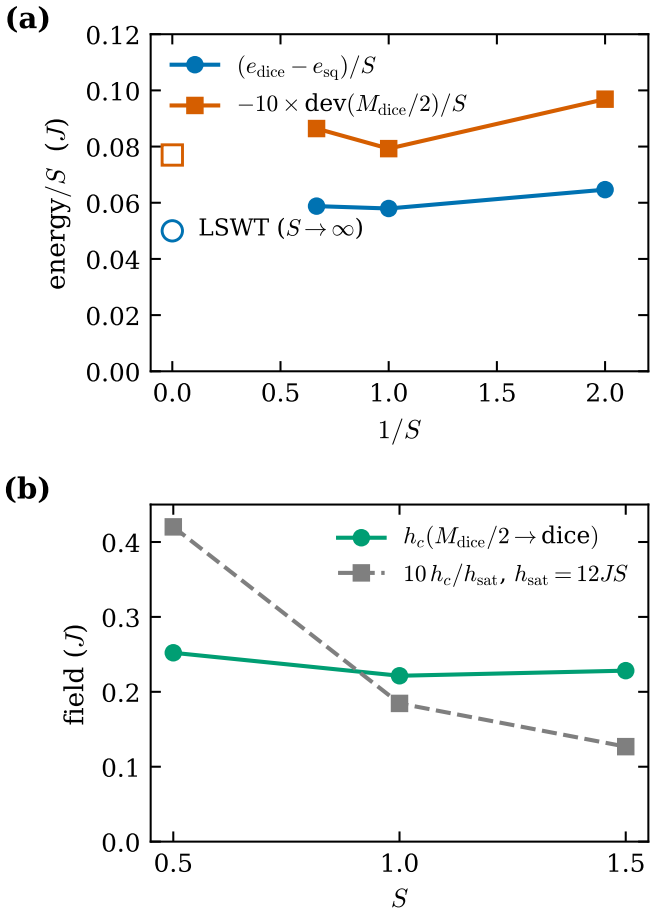



FIG. 9. Spin-$S$ QMC ($L = 12$, $\beta JS = 12$). (a) Selection energies per unit $S$: dice minus square-type (circles) and the $M_{\rm dice}/2$ deviation (squares, $\times(-10)$). Open symbols: LSWT ($S \to \infty$). Both approach finite limits, i.e., selection energies scale as $JS$. (b) The field at which the $M_{\rm dice}/2$ crystal gives way to dice at this system size (circles) is $S$-independent in units of $J$. Relative to the saturation scale, the intermediate region therefore shrinks as $1/S$ (squares).

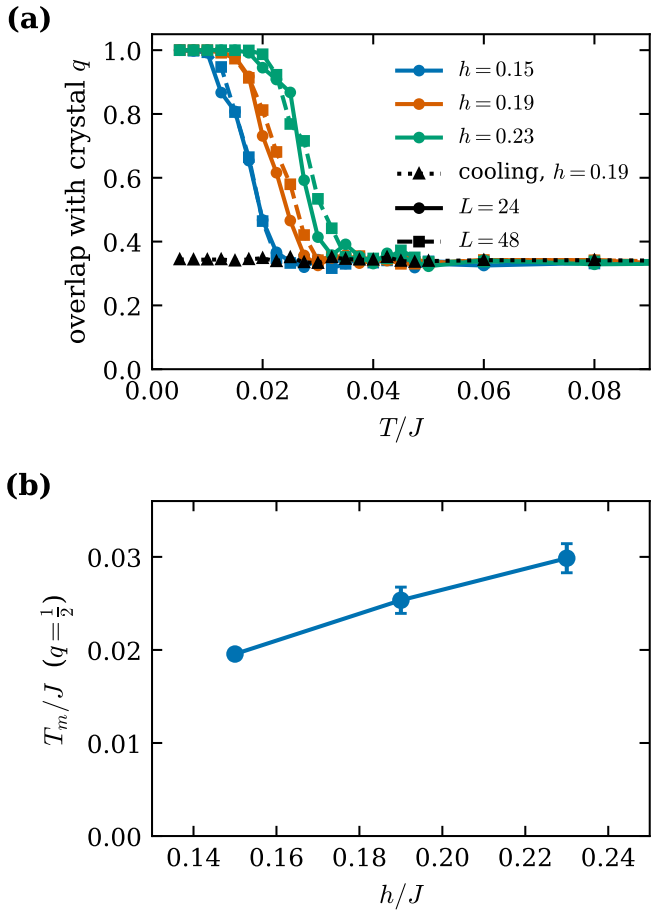



FIG. 10. Thermal stability of the $M_{\rm dice}/2$ crystal (tiling Monte Carlo with the QMC-scaled coordination model, local hexagon flips, crystal tilt sector). (a) Overlap $q$ of the tiling with the crystal on heating for three fields and two sizes, and on slow cooling from the disordered state (triangles). $q \simeq \frac{1}{3}$ is the value for an uncorrelated tiling. (b) Melting temperature, defined by $q = \frac{1}{2}$.

at the random value and the magnetization freezes near $0.47M_{\rm dice}$. Cooling 50 times more slowly does not change this [Fig. 12(c)]. The final state has $G = e_0 - hm$ ($S = \frac{1}{2}$) higher than the crystal by $0.9$–$1.3 \times 10^{-3}J$ per site. It is therefore a metastable state of the local-flip dynamics, i.e., kinetic trapping, and not evidence for an equilibrium glass phase. This trapping is a practical consideration for any realization in which the tiling is annealed.

Two omitted effects can be estimated. (i) *Magnon entropy.* From Bloch LSWT, the magnon free energy per site of the crystals at $T = 0.02$–$0.03J$ is $-1.6 \times 10^{-6}$ (square-type) to $-7\times10^{-5}J$ (dice), roughly proportional to $m$. On melting at $h = 0.19J$ the magnetization drops by $\Delta m \approx 0.018$ and the configurational entropy rises by $\Delta s \approx 0.1$ per site. The magnon correction shifts $T_m$ by about $\Delta F_{\rm mag}/\Delta s \sim 10^{-4}J$ and the step fields by $\lesssim 5\times10^{-4}J$, both negligible. (ii) *Tilt-changing moves* enlarge the configuration space of the disordered state but not of the locked crystal. They would therefore lower $T_m$, so our values are best read as upper estimates within this model. A joint treatment of spin and tiling fluctuations is left for future work.

## IX. GEOMETRIC INTERPRETATION: FOLDED AND CRUMPLED SURFACES

*Definitions.* Every lozenge tiling is the projection along [111] of a stepped surface $\Sigma$ made of unit squares of the cubic lattice [10]. We lift the tiling by assigning the steps $+\mathbf{e}_x$, $+\mathbf{e}_y$, $+\mathbf{e}_z$ to the bond directions $\mathbf{a}_1$, $\mathbf{a}_2 - \mathbf{a}_1$, $-\mathbf{a}_2$ (whose sum vanishes). Each lozenge becomes a unit square face, and the tiling graph becomes the edge skeleton of $\Sigma$. We verified that the lift is single-valued for all structures shown. Around a vertex of coordination $z_i$ there are exactly $z_i$ faces, each contributing a right angle,

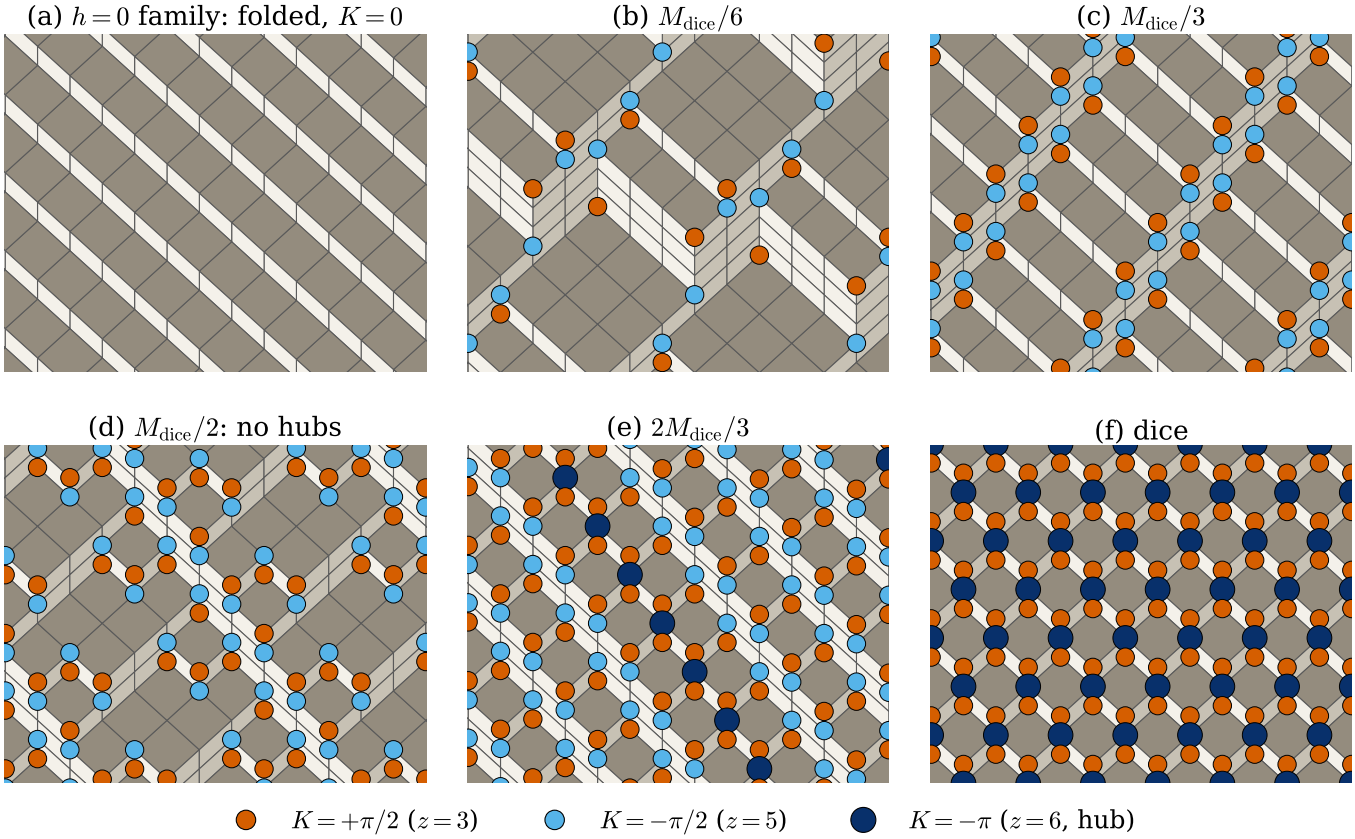



FIG. 11. Tilings lifted to stepped surfaces in $\mathbb{Z}^3$ (orthographic view, tilted by 12° from [111]; faces shaded by orientation). Colored dots mark vertices with nonzero discrete Gaussian curvature $K_i = 2\pi - \frac{\pi}{2} z_i$. (a) A zero-curvature tiling from the degenerate $h = 0$ family: two facets joined by straight folds, $K_i = 0$ everywhere. (b)–(e) The plateau crystals. Positive curvature ($z = 3$) sits only on the majority sublattice and negative curvature ($z = 5, 6$) only on the minority sublattice. The $M_{\rm dice}/2$ crystal (d) uses only $\pm\pi/2$ vertices. (f) Dice: $+\pi/2$ cones and $-\pi$ hubs.

so the discrete Gaussian curvature (angle deficit) is

$$K_i = 2\pi - \frac{\pi}{2} z_i = \tfrac{\pi}{2}(4 - z_i). \tag{6}$$

Sites with $z = 3, 4, 5, 6$ are cones ($+\pi/2$), flat points (0), saddles ($-\pi/2$) and strong saddles ($-\pi$). Two exact identities follow. First, $\langle z \rangle = 4$ is the discrete Gauss–Bonnet theorem on the torus, $\sum_i K_i = 0$. Second, inserting Eq. (6) into Eq. (4) gives

$$N_B - N_A = -\frac{1}{2\pi} \sum_i \eta_i K_i, \qquad S_{\rm tot} = \frac{S}{2\pi} \Big| \sum_i \eta_i K_i \Big|, \tag{7}$$

with $\eta_i = +1$ ($-1$) on sublattice $A$ ($B$). The total curvature always vanishes, and the Lieb moment is its sublattice-staggered part. We checked Eq. (7) numerically for every structure in Fig. 11. We distinguish this local quantity from the *tilt*, the average orientation of $\Sigma$. The tilt is fixed by the fractions of the three lozenge orientations and is conserved by hexagon flips.

*The results in this language.* (i) Because $z_i$ is linear in $K_i$, Eq. (2) states that the quantum zero-point energy is, to leading order, a convex function of the local Gaussian curvature. At zero field it is minimized by surfaces with $K_i = 0$ everywhere. These are not only the flat facet (the square-type tiling) but also facets joined by straight folds along cube axes [Fig. 11(a)], i.e., developable, origami-like surfaces. All of them are square lattices as graphs (every vertex has degree 4 and every face is a quadrilateral), so their Hamiltonians coincide in the bulk. Consistently, the 68 distinct zero-curvature crystals (cell, orientation) in our enumeration have thermodynamic-limit LSWT energies within $2.5 \times 10^{-6} J$ of each other, a spread attributable to the finite $k$ grid. Folding costs nothing; curvature does. (ii) By Eq. (7) the Zeeman energy $-hS_{\rm tot}$ couples to staggered curvature. The field therefore crumples the surface, placing cones on the majority sublattice and saddles on the minority one. Every crystal on the hull is *perfectly sorted*, $\sum_i |K_i| = |\sum_i \eta_i K_i|$: the mean $|K_i|$ is 20°, 40°, 60°, 80° and 120° for $M/M_{\rm dice} = \frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1$, i.e., exactly $120° \times M/M_{\rm dice}$. No curvature is wasted on the wrong sublattice. (iii) The hub-free kink of Sec. IV B reads: staggered curvature is cheapest in $\pm\pi/2$ vertices, $-\pi$ saddles cost 50% more per unit, and the $M_{\rm dice}/2$ crystal is the densest packing of $\pm\pi/2$ vertices we find. (iv) The

field does *not* control the tilt. The $M_{\rm dice}/6$, $M_{\rm dice}/3$, $M_{\rm dice}/2$ and dice surfaces all have the [111] orientation, whereas the $2M_{\rm dice}/3$ crystal is tilted by 25°, and the $h = 0$ family contains orientations from 35° to 55°. Bending of the average surface and crumpling are distinct, and only the latter carries the moment.

*Limitations of the picture.* (a) $\Sigma$ lives in an auxiliary height space. Its third dimension encodes connectivity and is not a physical displacement. Nothing here implies that a material buckles, unless the height is itself physical, which we do not assume. (b) The analogy with origami is exact only for the zero-field family, whose folds keep $K_i = 0$. The field-induced structures contain conical defects, so they are "crumpled with cones" rather than isometrically folded. There are no fold angles, rigidity or mechanics: all faces are unit squares with fixed right angles. (c) The energy is not an elastic bending energy. It is a sum of local curvature costs $\varepsilon(z_i)$ with pair corrections between curvature defects that decay within about three lattice spacings (Sec. V C), and there is no metric or bending modulus. The convexity is a numerical result of LSWT and QMC for $S = \frac{1}{2}$ to $\frac{3}{2}$, not a consequence of geometry. (d) Equations (6) and (7) and Gauss–Bonnet are exact. Statements about energies inherit the accuracy of the coordination model and of QMC discussed above. (e) The Legendre construction that turns $e_0(m)$ into the staircase is formally the same as the construction of faceted equilibrium crystal shapes. Here, however, $m$ is a staggered curvature density and not a surface orientation, so this analogy is structural only. (f) On the torus the tilt is fixed by the boundary conditions. The tilts quoted in (iv) are those of the optimal cells found and may be one of several nearly degenerate variants.

## X. DISCUSSION

*Height-model interpretation.* Section IX makes the height representation concrete: Eq. (2) is an effective height model whose potential is a convex function of the local Gaussian curvature of the lifted surface. The field acts as a chemical potential for the sublattice imbalance. In this language the plateaus are *locked* phases of the height model, commensurate crystals analogous to the ordered phases of quantum dimer models [15, 16]. The staircase is analogous to the devil's staircases of competing-interaction systems [17], but here the underlying potential is set by the convexity of $\varepsilon(z)$ and not by an imposed long-range interaction. Our data establish a finite set of steps, unchanged by 48-site cells. The short range of the defect interaction (Sec. V C) suggests that any further subdivision is confined to field windows of order $10^{-3}J$, but larger cells are needed to settle this.

*Relation to known plateaus.* Magnetization plateaus in quantum magnets usually arise from a crystal of magnetic excitations on a *fixed* lattice [18, 19]. Here the plateau is a crystal of the lattice itself, and the spin state on each plateau is the Lieb ground state of that lattice. The two mechanisms could coexist if bond strengths are also varied. Studying that would require going beyond the uniform-$J$ model (1).

*Platforms.* Three classes of system make the graph of exchange bonds a design or growth variable. (i) *$\pi$-magnetic nanographenes*, in which sublattice imbalance controls the ground-state spin in accordance with Lieb's theorem and the building blocks can be fused into extended chains and networks [20, 21]. (ii) *Atom-by-atom spin arrays* on surfaces, which realize arbitrary two-dimensional coupling graphs and allow local spectroscopy of spin excitations [22]. (iii) *Programmable quantum simulators*, such as Rydberg arrays, in which the connectivity is reconfigurable and antiferromagnetic correlations on hundreds of sites have been measured [23]. In (ii) and (iii) the energies $e_0(m)$ of different tilings can be compared directly, and the $M_{\rm dice}/2$ crystal is a concrete target structure. In (i) the step fields, 0.085–0.27$J$, and the melting temperature, 0.02–0.03$J$, give the relevant scales once $J$ is known. In small systems ($N \sim 10^2$) a further scale enters: a few uncompensated defects in a compensated background carry a size-dependent energy $\approx \alpha|\mu|/N$, with $\alpha \approx 3.4J$ for the staggered dice lattice in our LSWT, ED and QMC checks. This is comparable to the step fields at $N \sim 10^2$ and would shift the lowest steps in finite arrays.

*Limitations.* (a) The enumeration is exhaustive for cells of up to 48 sites (LSWT screening), and the hull structures were evaluated with QMC in cells of up to 36 sites; the 48-site cells add no hull point. Convergence beyond that, and for $2M_{\rm dice}/3$, which 48-site cells cannot host, is not demonstrated. Larger cells may add steps or shift boundaries. (b) The $M_{\rm dice}/3$ plateau rests on a margin of $0.47 \times 10^{-3}J$. With the structure used previously, $\delta e$ changed between $L = 24$ and $L = 36$, and the improved structure has been evaluated only at $L = 36$. That the earlier, histogram-based selection missed the best $M_{\rm dice}/6$ and $M_{\rm dice}/3$ structures shows that conclusions at the sub-$10^{-3}J$ level depend on the completeness of the search. (c) The coordination model is used only for screening. The defect lattice gas reaches $1.5 \times 10^{-5}J$, but it is fitted to LSWT, which underestimates the QMC deviations by a density-dependent factor of 1.25–1.5, and it covers only hub-free tilings. (d) The finite-temperature results refer to an effective tiling model at fixed tilt without thermal spin fluctuations. The quoted $T_m$ is a scale of that model, and the non-recrystallization is kinetic trapping. (e) Irregular networks were generated with a model-assisted annealing that is not exhaustive. Every irregular network we evaluated with QMC lies above the periodic hull.

## XI. CONCLUSION

We have shown that the classically degenerate family of lozenge-tiling networks is organized, quantum mechanically, by one simple rule: the zero-point energy is

a convex function of the site coordination. This rule selects the uniform square-type tiling at zero field, gives dice|square-type walls negative energy, and, in competition with the Zeeman energy of the Lieb moment, stabilizes, within the searched space, a staircase of periodic crystals rather than the classical jump or a disordered network. Because a moment is carried most cheaply by pairs of $z = 3$ and $z = 5$ sites, the densest hub-free tiling, the $M_{\rm dice}/2$ crystal, is a kink of the leading-order energy. It is the central new object. It is stable over $0.173J < h < 0.211J$ for $S = \frac{1}{2}$, within a staircase of $M_{\rm dice}/6$, $M_{\rm dice}/3$, $M_{\rm dice}/2$ and $2M_{\rm dice}/3$ crystals. Below $M_{\rm dice}/2$ the crystals are successive packing stages of $z = 3/z = 5$ dipoles, held in place by a short-ranged attraction between curvature defects of the same sign and a repulsion between defects of opposite sign. In an effective tiling model it melts at $T \simeq 0.02$–$0.03J$. Its field window is $S$-independent in units of $J$. Geometrically, zero field selects folded, curvature-free surfaces, and the field crumples them with sorted conical defects, because the Lieb moment is exactly the sublattice-staggered Gaussian curvature. These results identify quantum order by disorder acting on the geometry of the exchange network itself. They also provide concrete target structures and energy scales for nanographene networks, atom-by-atom spin arrays and quantum simulators, in which that geometry can be designed.

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### Appendix A: Tests of the coordination model, the wall picture and the cooling dynamics

*Coordination model.* The training set comprises the tilt family (dice, square-type, strips of width 6 and 12, and the superlattice of alternating six-row square-type and three-row dice strips, $s6d3$) at $L = 18$ and 24, each with 0 to 600 random hexagon flips, 128 tilings in total. The site model, Eq. (2), gives an rms residual of $1.55 \times 10^{-4}J$ ($1.60 \times 10^{-4}J$ leave-one-out). The largest single residual is $4.1 \times 10^{-4}J$. Residuals averaged over each class of starting structure and over each number of flips are below $1.8 \times 10^{-4}J$ and show no trend with the number of flips. Adding the densities of the ten bond types $(z_i, z_j)$ reduces the rms residual to $0.82 \times 10^{-4}J$ ($0.87 \times 10^{-4}J$ leave-one-out). The out-of-sample test on 71 periodic crystals [Fig. 12(a)] gives $2.4 \times 10^{-4}J$ for the site model and $1.5 \times 10^{-4}J$ with pair terms. Within groups of crystals that share the same $m$ and the same histogram, LSWT energies differ by up to $3.5 \times 10^{-4}J$ (at $M_{\rm dice}/3$) and $3.1 \times 10^{-4}J$ (at $M_{\rm dice}/2$). The Spearman rank correlation between the pair model and LSWT within each $m$ is 0.76–0.92. For the two $M_{\rm dice}/2$ structures evaluated with both methods, LSWT ($-0.64945$ vs $-0.64914J$) and QMC ($\delta e = -5.3(4)$ vs $-4.8(4) \times 10^{-3}J$ at $L = 24$) give the same order. For the two $M_{\rm dice}/3$ structures of the earlier search, LSWT separates them by $2.5 \times 10^{-4}J$. In QMC the LSWT-preferred one gives $-4.39(13) \times 10^{-3}J$ at $L = 36$ and $-3.6(3) \times 10^{-3}J$ at $L = 24$, while the other gives $-3.88(53) \times 10^{-3}J$ at $L = 18$; these values, at different sizes, do not establish the order. The class-resolved search later found $M_{\rm dice}/6$ and $M_{\rm dice}/3$ structures that are lower in LSWT by 2.5 and $1.1 \times 10^{-4}J$. In QMC at $L = 36$ they are lower by $0.45(15)$ and $0.14(16) \times 10^{-3}J$, from three and two independent runs. LSWT orderings at the level of a few $10^{-4}J$ are therefore borne out, whereas differences of order $10^{-4}J$ are not resolved by QMC.

*Walls.* Straight-wall superlattices at $m = 1/18$, $1/12$ and $1/9$ have LSWT energies $-0.65076$, $-0.64718$ and $-0.64246J$, compared with $-0.65281$, $-0.64945$ and $-0.64492J$ for the best crystals. Their histograms contain 6%, 8% and 17% hubs, respectively.

*Cooling.* Starting from the melted state at $T = 0.05J$ and cooling to $0.005J$ in 19 steps with 3000, 30 000 or 150 000 sweeps per step ($L = 24$, $h = 0.19J$) always ends with $q \approx 0.31$ and $G$ above the crystal by 0.9–$1.3 \times 10^{-3}J$.

### Appendix B: Spin-wave hull and finite-size effects

The LSWT energies of the annealed tilings relative to the linear interpolation are independent of system size for $L = 18$–30 (Fig. 13). This justifies the use of $L \leq 36$ tori. LSWT underestimates the QMC deviations of the hull crystals by factors of 1.5, 1.4, 1.3 and 1.25 at $M_{\rm dice}/6$, $M_{\rm dice}/3$, $M_{\rm dice}/2$ and $2M_{\rm dice}/3$, and the step fields by factors of 1.16–1.4. For periodic crystals with an extensive moment the LSWT energy per site on an $L = 36$ torus equals the thermodynamic limit to better than $5 \times 10^{-6}J$; no term proportional to the total moment is present (Sec. III). LSWT reproduces the set and order of the hull structures, but differences of order $10^{-4}J$ are not resolved by QMC (Appendix A). For periodic crystals we also evaluate LSWT directly in the thermodynamic limit by Bloch diagonalization of the supercell ($6 \times 6$ to $64 \times 64$ $k$ points). For the crystals checked it agrees with $L = 36$ tori to $10^{-8}J$. The $M_{\rm dice}/2$ crystal is the deepest LSWT hull point as well ($-3.9 \times 10^{-3}J$ in the thermodynamic limit).

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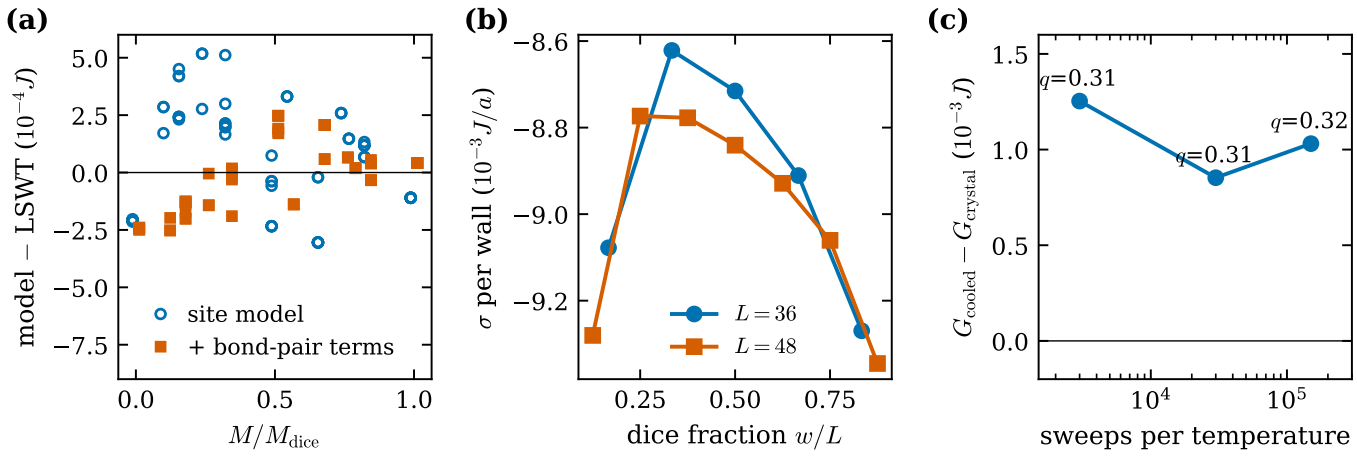



FIG. 12. (a) Out-of-sample test of Eq. (2) on 71 periodic crystals, relative to thermodynamic-limit LSWT: site model (circles) and site model with bond-pair terms (squares). Points are offset horizontally for clarity. (b) LSWT energy per straight dice|square wall for a dice strip of width $w$ in a square background on $L = 36$ and 48 tori. The walls are $w$ and $L - w$ rows apart, and closely spaced walls ($w/L$ near 0 or 1) have lower energy. (c) Free energy $G = e_0 - hm$ of the state reached by cooling to $T = 0.005J$ relative to the $M_{\rm dice}/2$ crystal, for three cooling rates; labels give the overlap $q$ with the crystal.

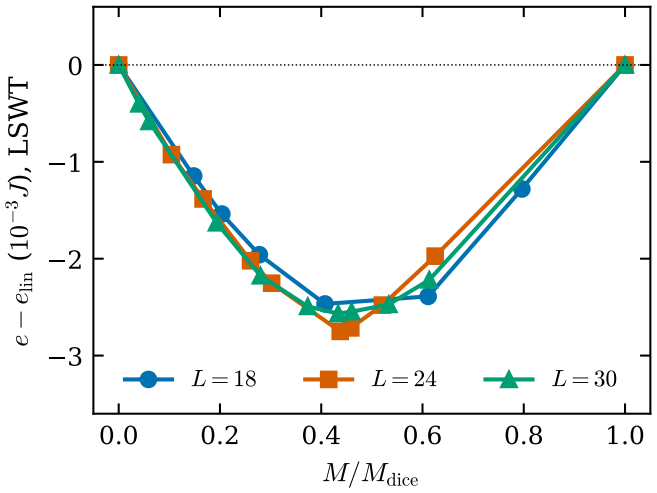



FIG. 13. LSWT energies of annealed tilings relative to the square-to-dice line for $L = 18$, 24, 30. The lower envelope is size-independent to within $3 \times 10^{-4}J$.


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